Identification of Sounds from Construction Sites: A Machine Learning Approach

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Abstract. The rate at which the world is evolving is astonishing with cutting-edge technologies being introduced every day. There have been developments in every field ranging from constructing gigantic architectures to enhancing the ease and comfort with which we can commute daily. Nevertheless, this has given rise to different types of pollution, sound pollution being one of them. Although the impact of sound pollution is not instantaneously noticed, it has a distinct effect on humans and nature which is observed over a period of time. There are multiple sources that cause sound pollution. Sounds from ongoing construction sites are one of them. Such sounds disrupt the daily lifestyle of people, their health, and daily productivity. Mostly, these sounds start very early in the day thereby making it difficult for people to manually take precautions like shutting doors and windows or deploying mobile soundproofing solutions. Machine Learning has advanced significantly over the years leading to the refinement of Artificial Intelligence. This has aided in the development of multifarious solutions to ease and simplify our daily lifestyle. This sets the stage for an autonomous system, that can detect sounds from construction sites very early to automatically deploy precautionary measures so that the effect of these sounds on close by residents can be minimized. In this paper, a system is presented that models audio clips using MFCC-based features for the identification of sounds from construction sites in the Urban environment. Experiments were performed on a dataset of over $6K$ clips composed of the varied construction sounds as well as ambient Urban sounds and a maximum accuracy of 98.02% was achieved using random forest-based classification.

Keywords: Sound pollution, Urbanized Landscape, Construction Sounds, Machine learning

1 Introduction

Over the years there have been remarkable advancements across all dimensions which has reduced hassle in the daily life of humans. Such advances span from medical treatment to communication as well as running regular errands. This evolution of technology has also contributed to the introduction of new modalities of pollution which is bringing about significant harm to our planet as well as to the health of all living creatures at an alarming pace. Among the varied types of pollution, sound/noise pollution is of high concern. While it may not exhibit any immediate consequence but acts as a camouflaged peril. It disrupts both the mental and physical health of humans [4]. There are various sources of sound pollution like vehicle sounds, loud microphones, firecrackers, etc. Sounds from construction sites are one such source. This is a global problem considering the fast-paced Worldwide urbanization, that has been catalyzed by advanced machines. Construction sounds are not just incidental sounds that are a result of development but they also exemplify the kernel of revolution and progression. The impact of construction sounds on the working productivity and contentment of places like schools, colleges, hospitals, libraries, and research centers cannot be undervalued. High levels of construction sounds near educational institutions disturb in-progress lectures and limit the overall academic performance of a student. Medical care institutions require a peaceful environment surrounding their location so high levels of construction sounds not only disturb the recuperation of the patient but also significantly compromise the efficacy of surgeries that demand undivided concentration. Systems that can automatically detect such sounds can be very helpful in this context. Such systems can aid in the automatic deployment of countermeasures like alarms, automatic shutting of windows, deployment of mobile soundproofing solutions, etc. to minimize the effect of noise on use. Such systems can also help in assessing the extent of construction noise in a region prior to procurement of properties. This can be very helpful for people who have serious health problems where such sounds can amplify their ailments.

2 Related works

There have been several initiatives for differentiating and identifying construction sounds. Moniruzzaman et al. [8]presented a system for evaluating the noise pollution levels of rush-hour hotspots in Khulna City, Bangladesh. The dataset encompassed location assessments of noise levels which were recorded at different times and gathered by means of a sound level meter(Multi-function Environmental Meter, Model: RH87). Various noise pollution indices were calculated such as Noise Climate(NC) and Traffic Noise Index(TNI) by the use of Gaussian percentiles. Inverse Distance Weighted(IDW) was used to develop noise maps that illustrated the noise level distributions in the study area. Shibbari intersection, North Kashipur and New market recorded highest noise levels of 83.2

[4] https://www.medicalnewstoday.com/articles/noise-pollution-health-effects#mental-health

dBA, 76.2 dBA and 79.6 dBA respectively. Surveys carried out among locals disclosed that noise levels in the industrial areas had harmful effects such as disturbance of sleep(36%), headaches(23%), hearing problems (21%), and irregular heartbeat(20%). Sojol et al. [14] built a two-tiered arrangement having the final goal of reducing the harmful effects of noise pollution on human health. The used dataset comprised data obtained from the Department of Environment in Bangladesh. The arrangement made use of Arduino Uno, Wifi modules, sound sensors, and microcontrollers. The first phase of the arrangement displayed sound levels and suitable memos to create awareness among the masses whereas the second phase dealt with regulating the noise levels by the use of a controlled honking system reducing unnecessary honking.

Angulo et al. [1] crafted an architecture aimed at evaluating the capabilities of the latest Urban sound event detection systems and analyzed the effect on the system's performance by factors like polyphony and sound proximity. The datasets used included SONYC-UST-v2 [2], SINGA:PURA [11] labeled set and SINGA:PURA unlabelled set. Convolutional Recurrent Neural Network(CRNN) and embedding classifiers with PaSST(Patchout faSt Spectrogram Transformer) and openL3 were utilized for the experimentation purpose. Best accuracy scores of 80.8% and 60.1% were achieved in the case of the CRNN system with the per-channel energy normalization (PCEN) [18] front-end. Ezhilarasi et al. [3] have proposed a system of monitoring air and noise pollution by the usage of embedded computing and sensory devices. The measured sound level was notified to the online server through the Internet of Things and the obtained data was made available to the end user via Wi-Fi.

Deshmukh et al. [5] put forward a system aimed at providing a versatile and cost-efficient system that specifically monitored air and sound pollution by means of the Internet of Things (IoT). The data was collected in real-time by sensors and was transmitted over the Internet with a Wi-Fi module(ESP8266). A microcontroller was also used for data processing. The system was tested at M.G Road in Pune where observed values of carbon monoxide, carbon dioxide, and sound levels were recorded as 44%, 48%, and 36% respectively. The results achieved were near to that obtained by standard devices exhibiting the efficiency of the system. Tsao et al. [16] presented a system to measure and better envision noise pollution in expanded environments by applying a distributed sound sensing system (DSSS) employing the Internet of Things. An Ameba RTL 8195AM as well as a microphone module was used for the construction of the DSSS while C programming language was used for the development of the system's firmware. Also, PHP was used for the development of a clouding platform that would receive data from several DSSS devices. The system was been successfully set up in the library of National Quemoy University in Kimmen.

Kumbalkar et al. [6] created a system that provided real-time data on air and sound quality. The system included components such as a Node MCU controller, air sensors, Wi-Fi and sound detector module, LCD display, and a mobile application allowing users to track pollution data on cell phones. Alerts are initiated when pollution surpassed preset levels. Saha et al. [13] created a framework that

could monitor temperature, humidity as well and air and sound pollution levels. The data was collected in different locations through the LM393 sound sensor, MQ135 gas sensor, and the DHT11 temperature humidity sensor. The sensors were integrated with the Raspberry Pi acting as a core processing unit from where the collected data was transferred for deeper analysis and supervision. The framework successfully presented a consistent tracking of the parameters and identified any unusual or hazardous conditions. Manglani et al. [7] developed a system that consistently kept track of the air and noise pollution levels in an area. The sensors collected the data measurements of gases such as NH_3 and CO_2 and the sound intensity levels in decibels. Air sensor MQ135 and noise sensor were integrated with an Arduino microcontroller. An alert was activated, indicated by a buzzer when the predefined limits of pollution were crossed. Rafi et al. [12] presented a system for the assessment of sound levels in a city. The sound level data were collected spanning over 30 distinct locations in Dhaka, Bangladesh by the use of the Lutron SL-4012 sound level meter during peak traffic hours. Further, a sound pollution heatmap was provided which portrayed a categorization of the recorded sound levels as per the World Health Organization's standards over Dhaka. Additionally, a device-based sound metering and awareness messaging system was proposed which could contribute to raising public awareness and reducing sound pollution.

3 Dataset Description

The dataset encompasses diverse sounds that have been sorted into two categories. These categories aid in illustrating and depicting the sophisticated array of noises that are present in Urban. The first category consists of construction sounds that have become an integral part of our daily lives and are an essential part of the Urban fabric. This category consisted of different sounds like heavy machinery sounds, drilling sound, concrete mixing sound, backup generator sound, safety whistles sound, compaction and hydraulic equipment sounds, welding sounds, hammering and nailing sounds, clanging of metal sound, sawing sound, demolition sounds, road paving sound, tunneling and boring sounds. The second category consisted of an extensive selection of additional ambient sounds ordinarily experienced in a metropolitan atmosphere. These sounds consisted of both indoor and outdoor sounds like car honks, animal sounds, kitchen sounds, conversation, rain and thunder sounds, etc. Both the classes (Construction and Rest) were composed of these sounds with clips of 5 seconds. The construction class consisted of 3179 clips while the rest class consisted of 2896 clips. It was noted that both classes had commonalities in terms of ambient noise, conversations, airplane sounds, etc. Such characteristics were not removed to ensure real-world conditions. The word cloud of the constituting sounds of both classes is presented in Figure 1.

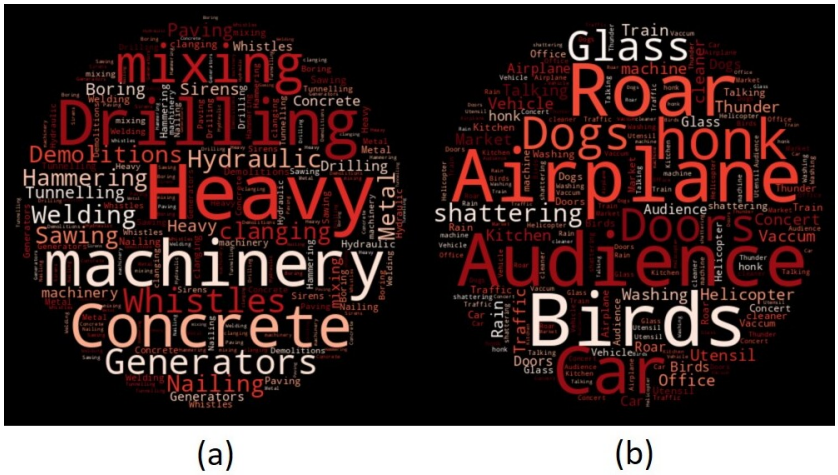

Fig. 1. Wordcloud of constituent sounds of (a) Construction class, (b) Rest class.

4 Proposed Method

4.1 Feature Extraction

An audio demonstrates high variation throughout its length which poses difficulty in analysis. To tackle this problem, the audio is split into shorter sub-clips (frames) to ensure pseudo-stationarity. At times, framing leads to loss of continuity which leads to jitters during analysis in the frequency domain. Thus, the frames are subjected to windowing to address this issue. In this experiment, Hanning window [15] was used for this purpose which is illustrated as follows:

$$w(n) = 0.5 * \left(1 - cos\left(2\pi \frac{n}{N}\right)\right) \tag{1}$$

where length of window $= N + 1$ and $0 \leq n \leq N$.

The audio clips were split into frames of different lengths from 512 samples to 1792 samples. Standard Mel Frequency Cepstral Coefficients (MFCC) were extracted for each of these clips at frame level. MFCC involves the use of the Mel-frequency scale that is based on the idea of how humans recognize and understand the variances in frequencies of sound [9]. The audio signal is converted from the Hertz (Hz) scale to the Mel scale. The Mel scale corresponds to how humans perceive sound. The relationship between the Hertz and Mel scale is represented as:

$$m = 1127 * \log\left(1 + \frac{f}{700}\right) \tag{2}$$

where m symbolizes the value on the Mel scale and f expresses the frequency in Hertz. A filter bank of 20 Mel Filters in the range of 20Hz to 20000 Hz is presented in Figure 2.

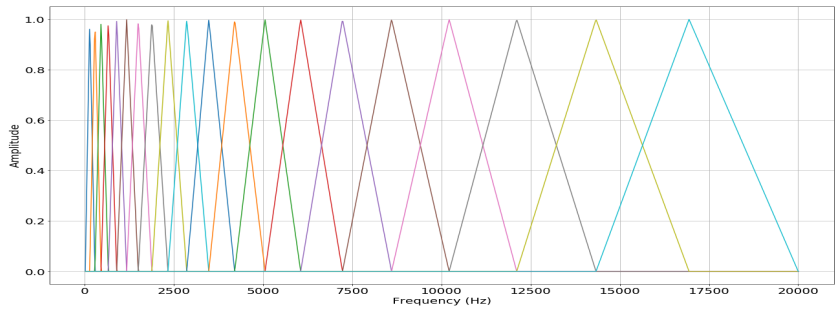

Fig. 2. Filter bank of 20 Mel filters.

4.2 Machine Learning-based Classification

Machine Learning can be referred to as the branch of Artificial Intelligence(AI) that is particularly inspired by the human brain to learn and evolve from experiences. It is concerned with the development and deployment of algorithms that make use of pre-existing data also known as training data to generate output. The output thus obtained helps in making predictions and decisions on test data. An increase in the amount of training data leads to an increase in the output efficiency of the machine learning model in most cases.

Ensemble learning is a type of Machine Learning that merges the results obtained from several individual models commonly of identical or diverse types to develop a more potent and enhanced predictive model. The Random Forest is a classifier that is based on this technique. Random Forests are used in both regression and classification tasks. The root concept of Random Forest lies in the fact of building up several decision trees and merging the predictions obtained to make the results more accurate and efficient. It can be used to calculate the significance of each feature and is capable of handling absent values in the dataset. Over-fitting is minimized in Random Forests as irregularity/randomness is incorporated at the time of data sampling as well as during feature selection.

The Random forest algorithm includes the following steps:

- **Ensemble of decision trees**: A group of decision trees is created that work in collaboration with each other. Each of these trees contributes towards making predictions.
- **Random sampling**: In the course of training different random samples of the data are taken which is then followed by choosing certain data points and utilizing them to build a tree. Each tree is trained on a set of samples.
- **Random selection of feature**: During the construction of each decision tree only a number of features are chosen from all the available features. This contributes to making the constructed trees distinct from each other.
- **Classification and Regression**: In classification, the final predicted outcome is the most typical prediction from all the trees. In regression, the final predicted outcome is the average of all the tree predictions.

– **Enhanced accuracy**: The generated decision trees are combined, using random samples and random features being selected. This reduces over-fitting and contributes to a Random Forest's ability to handle noisy data.

5 Results and analysis

Each of the feature sets was subjected to the random forest-based classifier. 70% of the data was used for training while the remaining was used for testing. The obtained accuracies for the disparate feature sets are presented in Table 1.

Table 1. Performance of different feature sets. The columns represent the window lengths while the rows represent the feature dimensions.

	512	768	1024	1280	1536	1792
5	95.2	95.09	95.2	95.37	95.64	95.64
10	96.97	97.24	97.52	97.68	97.46	97.79
15	96.91	96.97	97.41	97.68	97.74	97.74
20	96.91	96.91	97.02	97.35	97.68	97.68
25	96.69	96.86	97.46	**97.85**	97.63	97.57
30	96.8	97.3	97.35	97.52	97.57	97.68

It is noted that the best performance was obtained for the 25 dimensional features computed using a window length of 1280 samples. The confusion matrix for this setup is presented in Table 2.

Table 2. Confusion matrix for random forest with 50 estimators on the 25 dimensional features computed using a window length of 1280 samples.

	Construction	Rest
Construction	978	17
Rest	22	797

Next, the number of estimators varied from 10 to 100, and a higher accuracy of 98.01 % was obtained for both 35 and 36 estimators. The variation of accuracy for the different estimators is presented in Figure 3.

The confusion matrix for this setup is presented in Table 3. In this setup, an improvement of 0.17% was observed in the overall accuracy. The number of misclassifications was reduced by 1 and 2 for construction sounds and other sounds respectively. The misclassified instances were analyzed in detail and it was noted that some of the clips in construction sound had similarities with different indoor sounds like mixer grinder, oven alarms, forceful door closing, etc. Certain construction sounds also had distinct conversations which were also misclassified.

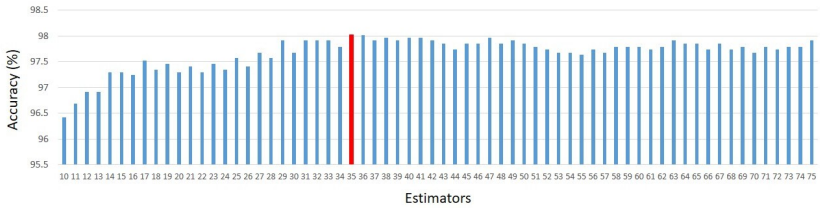

Fig. 3. Performance for the different number of estimators.

Table 3. Confusion matrix for random forest with 50 estimators on the 25 dimensional features computed using a window length of 1280 samples.

	Construction	Rest
Construction	979	16
Rest	20	799

The performance of several other classification techniques namely Naive Bayes [19], SVM [17], KNN [4], Decision Tree [10], and ADABoost [20] was also compared on the best-performing features whose results are presented in Figure 4. Both in terms of the average accuracy and the other sound class accuracies, random forest held the top spot. In terms of the construction sound class KNN produced a slightly higher accuracy of 98.79% which was 0.0.41% higher than random forest. However, in the case of other sounds, the recognition rate significantly dropped by 5.14% as compared to random forest. This was followed by SVM, where a reduction of 0.1% and 0.99% was observed for the construction and other sounds respectively. The lowest performance was obtained for Naive Bayes which was almost 16.71% less than that of Random Forest in terms of the average accuracy. The reduction of recognition rate was higher for the other sound class as compared to the construction sound class.

Fig. 4. Performance for the different classifiers.

6 Conclusion

In this paper, a system is presented to distinguish construction site sounds from acoustic surroundings in the urbanized landscape. Experimentation was conducted with construction sounds along with an assortment of familiar sounds occurring in the urbanized landscape. The system was successful in distinguishing the 2 classes with an accuracy of 98.02% using MFCC-based features coupled with Random Forest-based classification. In subsequent time, the framework will be evaluated on a dataset of increased size and variety in terms of the type of sounds. The dataset will also be exposed to additional noise to test the reliability of the framework. Tests will be carried out with manually designed attributes for enhanced productivity in modeling the audios to build a handcrafted feature-based architecture. The architecture will be developed further to analyze as well as identify subtle distinctions among various kinds of construction sounds. Eventually, the machine learning-based architecture will be enhanced to develop an airier framework to ease the deployment process in settings with a limited number of resources which will additionally contribute to instantaneous and uninterrupted computation. Finally, deep learning-based approaches will also be explored to enhance the efficacy of the proposed system.

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